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கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2016 ஓகஸ்ற்
General Certificate of Education (Adv. Level) Examination, August 2016

භෞතික විද්‍යාව I
பௌதிகவியல் I
Physics I

01 E I

පැය දෙකයි
இரண்டு மணித்தியாலம்
Two hours

Instructions:

* *This question paper consists of* **50** *questions in* **10** *pages.*
* *Answer* ***all*** *the questions.*
* *Write your* ***Index Number*** *in the space provided in the answer sheet.*
* *Read the instructions given on the back of the answer sheet carefully.*
* *In each of the questions* **1** *to* **50**, *pick one of the alternatives from (1), (2), (3), (4), (5) which is* ***correct*** *or* ***most appropriate*** *and* ***mark your response on the answer sheet with a cross*** (×) ***in accordance with the instructions given on the back of the answer sheet.***

Use of calculators is not allowed.

(Acceleration due to gravity, $g = 10\,\mathrm{N\,kg^{-1}}$)

1. The SI unit used to measure the activity of a radioactive source is
(1) Bq (2) Gy (3) J $\mathrm{Bq^{-1}}$ (4) $\mathrm{Bq^{-1}}$ (5) Sv

2. The percentage error of a certain length measurement has to be kept below 1%. If the error due to the measuring instrument is 1 mm, the measuring length has to be greater than
(1) 1 mm (2) 1 cm (3) 10 cm (4) 1 m (5) 10 m

3. A certain liquid-in-glass thermometer with a uniform bore radius has been calibrated using the boiling point of water and the melting point of ice. Of the following properties, what is the **most essential** property that a thermometric liquid used in this thermometer must possess?
(1) high volume expansivity (2) uniform volume expansion
(3) high thermal conductivity (4) low specific heat capacity
(5) low vapour pressure

4. Which of the following is **not true** regarding electromagnetic waves?
(1) Directions of electric and magnetic fields are perpendicular to each other.
(2) Speed does not depend on the medium of propagation.
(3) Do not necessarily require a material medium for propagation.
(4) Direction of propagation of the wave is perpendicular to the directions of electric and magnetic fields.
(5) Can be reflected at the boundary between two media.

5. A student has suggested the following three methods (A), (B) and (C) to increase the voltage sensitivity (V/cm) of a potentiometer wire.
(A) Increasing the length of the wire
(B) Connecting a resistor in series with the wire
(C) Increasing the voltage applied across the wire
Of the above three methods,
(1) only A is correct. (2) only A and B are correct.
(3) only B and C are correct. (4) only A and C are correct.
(5) all A, B and C are correct.

6. In a certain transformer there are 360 turns in the primary coil and 30 turns in the secondary coil. Which of the following voltage conversions is done using this transformer?
(AC = Alternating current, DC = Direct current)
(1) 240 V AC voltage to 12 V DC voltage.
(2) 240 V AC voltage to 2 880 V AC voltage.
(3) 240 V DC voltage to 20 V DC voltage.
(4) 240 V AC voltage to 20 V AC voltage.
(5) 240 V DC voltage to 2 880 V DC voltage.

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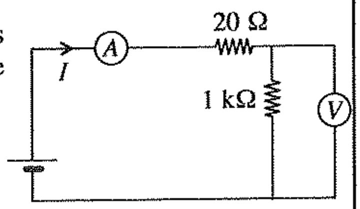
7. Of the following sets of internal resistances given, the set of internal resistances that suits best for an ammeter (A) and a voltmeter (V) to have in order to measure the current I and the voltage across 1 kΩ resistor of the circuit shown is



	Internal resistance of ammeter	Internal resistance of voltmeter
(1)	1 Ω	5 kΩ
(2)	5 Ω	1 kΩ
(3)	1 Ω	20 Ω
(4)	20 Ω	5 kΩ
(5)	5 Ω	50 Ω

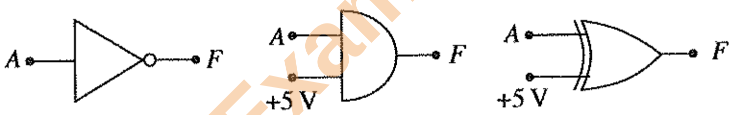
8. Which of the following is **not** a result of surface tension?
(1) Formation of spherical water droplets
(2) Capillary rise of water
(3) Ability of insects to walk on water surfaces without sinking
(4) The excess pressure inside a soap bubble
(5) Escaping of water molecules from water surfaces

9. Consider the following statements made about a standing wave on a stretched string.
(A) The energy does not propagate along the string.
(B) The position of a node does not vary with time.
(C) Maximum displacement achieved by each particle in the string depends on its position along the string.

Of the above statements,
(1) only A is true.
(2) only B is true.
(3) only A and C are true.
(4) only B and C are true.
(5) all A, B and C are true.

10. Which of the following gates operate/s according to the truth table given?

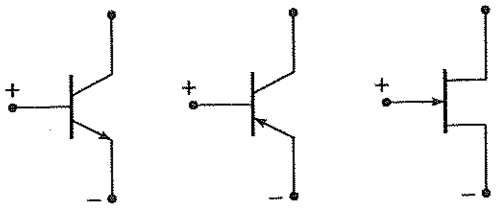
A	F
0	1
1	0



(P) (Q) (R)

(1) P only
(2) P and Q only
(3) Q and R only
(4) P and R only
(5) All P, Q and R

11. Which of the figures shown correctly indicate/s the polarities of potential difference that have to be applied across the junctions shown in order to operate the transistor properly and obtain a suitable current?
(1) in A only
(2) in B only
(3) in C only
(4) in A and C only
(5) in B and C only



(A) (B) (C)

12. When the body temperature of a person is 35 °C, the peak wavelength of the radiation emitted from the body occurs at 9.4 μm. If his body temperature increases to 39 °C, the peak wavelength will be (Assume that the black body radiation conditions can be applied)

(1) $\frac{35}{39} \times 9.4$ μm
(2) $\frac{39}{35} \times 9.4$ μm
(3) $\frac{77}{78} \times 9.4$ μm
(4) $\frac{78}{77} \times 9.4$ μm
(5) $\left(\frac{78}{77}\right)^4 \times 9.4$ μm

13. A moving jet plane can create a maximum sound intensity level of 150 dB. Take the sound intensity at the threshold of hearing as 10^{-12} W m^{-2}. The maximum intensity of the sound that can be created by the jet plane in W m^{-2} is
(1) 100
(2) 200
(3) 400
(4) 800
(5) 1 000

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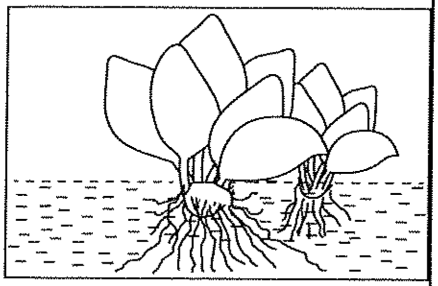
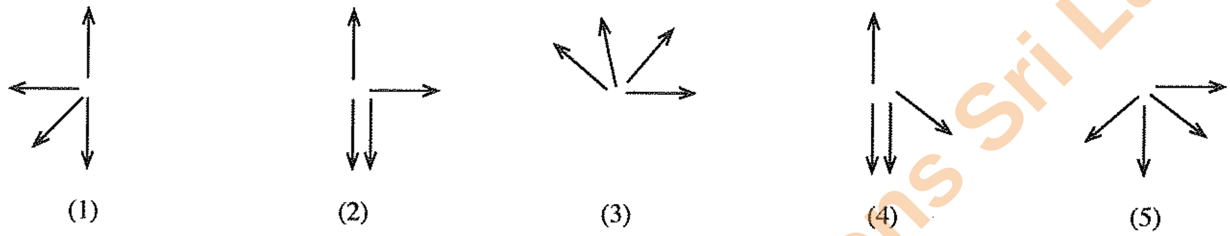
14. When wind blows over the surface of a still lake, a bunch of water hyacinth floating on water as shown in figure is observed to move in the direction of the wind with a velocity v. Consider the following statements made about v.



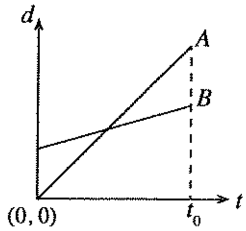
(A) Magnitude of v depends on the rate at which the momentum is transferred from air molecules to the bunch.
(B) Magnitude of v depends on the viscosity of water.
(C) Magnitude of v depends on the mass of the bunch.

Of the above statements,
(1) only C is true. (2) only A and B are true.
(3) only B and C are true. (4) only A and C are true.
(5) all A, B and C are true.

15. An object falling down vertically in air suddenly explodes into four pieces. Which of the following diagrams shows the possible **directions** of motion of the pieces immediately after the explosion? (↓ - direction of the object before explosion)



16. The two straight lines shown in the displacement (d) – time (t) graph represent the motions of two objects A and B started from rest at time $t = 0$ and moving along the positive x-direction. Which of the following statements made about the motions of the objects is true?

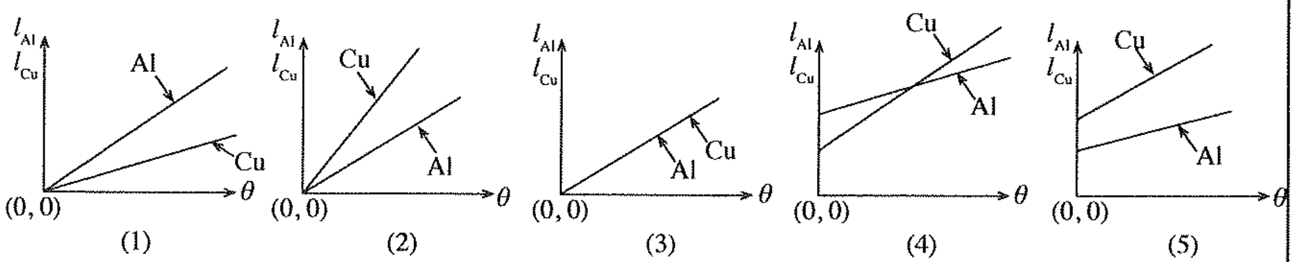


(1) The object A has travelled for a longer time than B.
(2) When $t = t_0$ object B has made a displacement greater than A.
(3) Object A has a greater velocity than B.
(4) Object A has a greater acceleration than B.
(5) Both objects have the same velocity at the point where the two straight lines cross each other.

17. An elevator of weight 5000 N carries a load of 5 000 N. While moving vertically upwards in a building, it travels at constant velocity from 2nd floor to 12th floor in 20 seconds. The height of each floor is 4 m. If only 80% of the power generated by the motor is consumed to lift the elevator and the load against gravity while moving at constant velocity, the power of the motor is
(1) 20 kW (2) 25 kW (3) 40 kW (4) 60 kW (5) 1 000 kW

18. Three monochromatic light beams A, B and C have the same intensities (i.e. energy flow through unit area per second). However, the wavelength associated with beam A is longer than that of B, and the frequency associated with beam C is smaller than that of A. The photon flux (number of photons crossing a unit area per second) of three beams when written in the ascending order, it will be
(1) C, A, B (2) B, A, C (3) A, B, C (4) B, C, A (5) C, B, A

19. l_{Al} and l_{Cu} respectively represent **fractional increase** in the original lengths of two rods of aluminium (Al) and copper (Cu) when their temperature is increased by an amount of θ °C from the room temperature. Which of the following graphs best represents the variations of l_{Al} and l_{Cu} with θ °C? (Linear expansivities of aluminium and copper are 2.3×10^{-5} °C^{-1} and 1.7×10^{-5} °C^{-1} respectively.)



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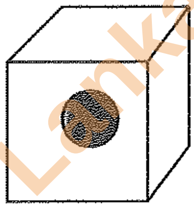
20. During the recent hot season, the night time temperature of a certain room with closed windows in a house made of bricks was observed to be 35 °C. A person opened the windows of the room for a few minutes at night and allowed the room to be filled with cooler air at 27 °C which was present outside the house. Once the windows were closed again, he observed that the temperature of the room had returned almost to 35 °C in a quick time. Which of the following reasons he had proposed to explain the observed effect is most **unlikely** to be accepted?

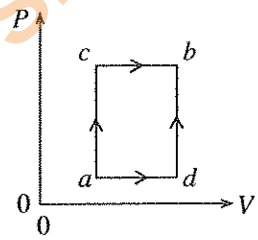
(1) Rapid movement of air molecules inside the room
(2) Collision of air molecules with the walls
(3) Low specific heat capacity of air
(4) Low thermal conductivity of air
(5) High specific heat capacity of brick walls

21. A cube of ice of mass 1 kg at 0 °C has a small metal sphere trapped inside as shown in the figure. It was found that this ice cube requires 300 kJ of heat energy to completely melt and form water at 0 °C. Specific latent heat of fusion of ice is 330 kJ/kg. The mass of the metal sphere in grams is approximately



(1) 30 (2) 33 (3) 91
(4) 110 (5) 333

22. An ideal gas is taken from state a to state b through two paths acb and adb as shown in the $P - V$ diagram. When going through path acb, 100 J of heat is absorbed and 50 J of work is done by the gas. If the work done by the gas, when taking the path adb is 10 J, the amount of heat absorbed by the gas during the path adb is



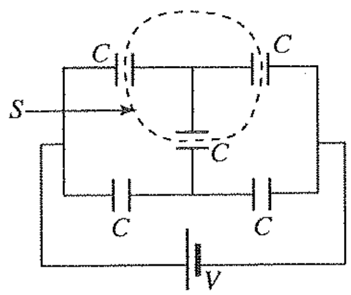
(1) 40 J (2) 50 J (3) -50 J
(4) 60 J (5) -60 J

23. If the ratio, $\dfrac{\text{mass of the planet}}{\text{radius of the planet}}$ for planet A is four times that of planet B, then the ratio

$\dfrac{\text{Escape velocity at the surface of Planet } A}{\text{Escape velocity at the surface of Planet } B}$ is

(1) $\sqrt{2}$ (2) 2 (3) 4 (4) 8 (5) 12

24. A network consisting of five identical parallel plate capacitors of capacitance C each, is connected to a cell of voltage V as shown in the figure. Assume that the capacitor plates are in free space. The net electric flux through the enclosed surface S is

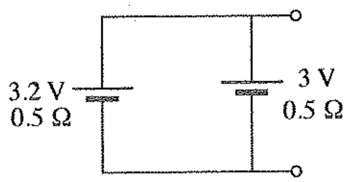


(1) $\dfrac{CV}{2\epsilon_0}$ (2) $\dfrac{3CV}{5\epsilon_0}$ (3) $\dfrac{CV}{\epsilon_0}$

(4) $\dfrac{3CV}{\epsilon_0}$ (5) 0

25. Two cells having e.m.f.s of 3 V and 3.2 V and equal internal resistances of 0.5 Ω are connected in parallel as shown in figure.

Power dissipation by the cell combination is



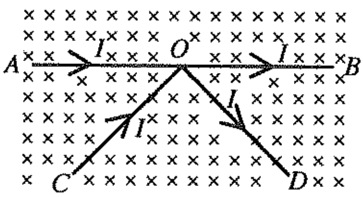
(1) 0.01 W (2) 0.02 W (3) 0.03 W
(4) 0.04 W (5) 0.05 W

26. Nine identical wires made of a certain metal, each of diameter d and length L, are connected in parallel to form a single resistor. The resistance of this resistor is equal to the resistance of a single wire of length L and diameter D made of the same metal if D is equal to

(1) $\dfrac{d}{3}$ (2) $3d$ (3) $6d$ (4) $9d$ (5) $18d$

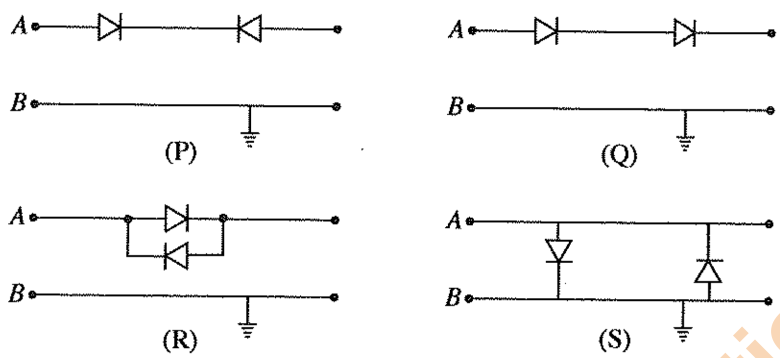
[See page five

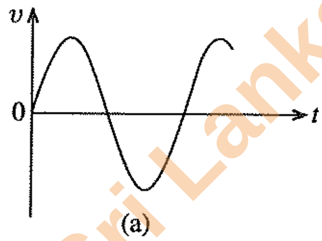
27. A structure consisting of straight wire sections of AO, OB, CO and OD of equal lengths arranged so that $A\hat{O}C = B\hat{O}D$, carry currents I along the directions shown. When this structure is placed perpendicular to a magnetic field as shown in the figure, due to magnetic field it will experience



(1) a resultant force along the plane of the paper in the upward direction.
(2) a resultant force along the plane of the paper in the downward direction.
(3) a resultant force along the plane of the paper to the right.
(4) a resultant force along the plane of the paper to the left.
(5) no resultant force.

28. The waveform shown in figure (a) is applied across the input terminals A, B of the circuits P, Q, R and S shown below.

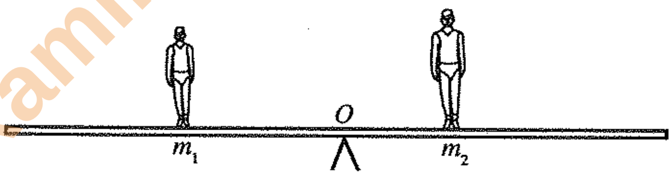




(a)

If the potential drops across the diodes are negligible, the input waveform will travel unaffected through
(1) the circuit P only. (2) the circuit Q only. (3) the circuit R only.
(4) the circuit S only. (5) the circuits R and S only.

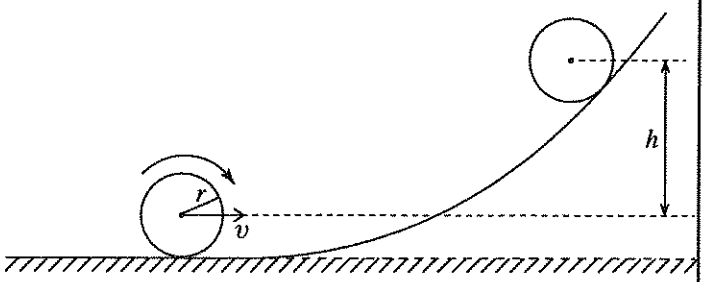
29. Two children of masses m_1 and m_2 are standing in equilibrium as shown in figure, on a uniform rod which is balanced at its centre of gravity O. Then they start moving simultaneously on the rod at constant speeds v_1 and v_2 respectively while maintaining the horizontal equilibrium of the rod. Consider the following statements made about the motion of the two children.



For the equilibrium to be maintained at any time t,
(A) they should always move in opposite directions.
(B) they should move keeping their total linear momentum always equal to zero.
(C) they should move so that the moment produced by one child about O is always equal and opposite to the moment produced by the other child about O.

Of the above statements,
(1) only A is true. (2) only B is true.
(3) only A and B are true. (4) only B and C are true.
(5) all A, B and C are true.

30. A uniform disc of mass m and radius r rolls without slipping, initially along a horizontal surface, and subsequently starts to climb up a ramp as shown in the figure. The disc has a linear velocity v on the horizontal surface. The moment of inertia of the disc about the axis through its centre and normal to the plane of the disc is $\frac{mr^2}{2}$. What is the maximum height h to which the centre of mass of the disc climb?



(1) $\frac{v^2}{2g}$ (2) $\frac{3v^2}{2g}$ (3) $\frac{3v^2}{4g}$ (4) $\frac{v^2}{g}$ (5) $\frac{2v^2}{g}$

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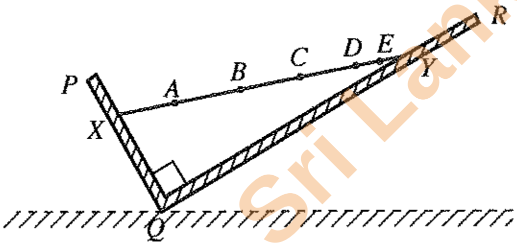
31. A glass of fresh orange solution of volume 500 cm^3 contains a few orange seeds at its bottom. It was observed that the seeds just began to float at the bottom when 10 grams of sugar was dissolved in the solution. Assume that the addition of sugar does not alter the volume of the solution. If the density of the orange solution before adding sugar was 1 000 kg m^{-3}, the density of orange seeds (in kg m^{-3}) is approximately equal to

(1) 1 020 (2) 1 040 (3) 1 060 (4) 1 080 (5) 1 100

32. A boy, sitting on a smooth turntable with a weight in his each extended hand, is rotating with an angular velocity ω_0. When he bends his hands towards his body, the angular velocity becomes ω_1. If I_0 and I_1 are the moments of inertia of rotating systems when the hands are extended, and bent towards his body respectively, then

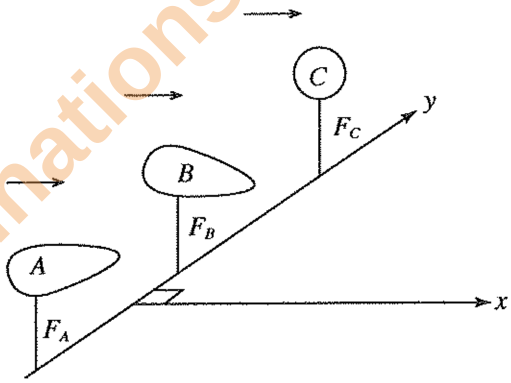
(1) $\omega_0 > \omega_1$, $I_0 > I_1$ and $\omega_0 I_0 > \omega_1 I_1$ (2) $\omega_0 < \omega_1$, $I_0 > I_1$ and $\omega_0 I_0 < \omega_1 I_1$
(3) $\omega_0 < \omega_1$, $I_0 > I_1$ and $\omega_0 I_0 = \omega_1 I_1$ (4) $\omega_0 > \omega_1$, $I_0 < I_1$ and $\omega_0 I_0 = \omega_1 I_1$
(5) $\omega_0 = \omega_1$, $I_0 = I_1$ and $\omega_0 I_0 = \omega_1 I_1$

33. A rod XY rests between two smooth boards PQ and QR kept inclined to the horizontal as shown in the figure. Angle PQR is 90° and the surfaces of the boards are normal to the plane of the paper. The centre of gravity of the rod is most likely to be situated at the point



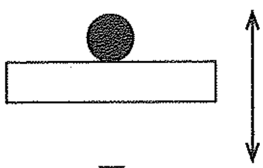
(1) A (2) B (3) C
(4) D (5) E

34. Two objects A and B of the shapes shown in the figure, and a spherical object C, all having identical masses, are mounted rigidly on a horizontal surface along the y–axis by three thin rods as shown in the figure. Both x and y axes are located on the horizontal surface.
A stream of air flows through the objects parallel to the surface and along x-direction. (Assume that the air flow causes no turbulence around the objects.) The magnitudes of the forces F_A, F_B and F_C exerted by the objects and the sphere on the mounted rods, when written in the ascending order, it will be



(1) F_B, F_A, F_C (2) F_B, F_C, F_A (3) F_C, F_A, F_B
(4) F_A, F_C, F_B (5) F_C, F_B, F_A

35. A mass is resting on a horizontal surface which moves up and down performing simple harmonic motion with amplitude A as shown in figure. The maximum frequency with which the surface can move while keeping the mass always in contact with the surface is

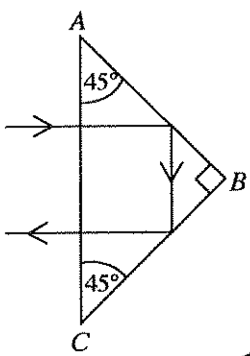


(1) $2\pi\sqrt{\frac{g}{A}}$ (2) $\sqrt{\frac{g}{A}}$ (3) $\frac{1}{2}\sqrt{\frac{g}{A}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{g}{A}}$ (5) $\frac{1}{\pi}\sqrt{\frac{g}{A}}$

36. A whistle emitting a sound of frequency f moves along the circumference of a circle of radius r at a constant angular velocity ω. v is the velocity of sound in air. The highest frequency of sound heard by a listener, who is at rest outside the circle is

(1) $f\left(\frac{v}{v-r\omega}\right)$ (2) $f\left(\frac{v-r\omega}{v}\right)$ (3) $f\left(1-\frac{v}{r\omega}\right)$ (4) $f\left(\frac{v}{r\omega}\right)$ (5) $f\left(\frac{v}{v+r\omega}\right)$

37. A ray of light is incident perpendicular to the surface AC of a right angled glass prism as shown in the figure. Minimum value of the refractive index of the material of the prism for which the ray will follow the path shown is



(1) 1.22 (2) 1.41 (3) 1.58
(4) 1.73 (5) 1.87

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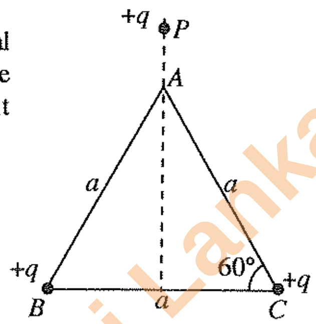
38. When an object is placed on the principal axis of a thin convex lens of focal length f_1, it forms a real image at a distance V_1 with a linear magnification of m_1. When this lens is replaced by another thin convex lens of focal length f_2, $(f_2 < f_1)$, being kept at the same position the new image distance V_2 and the magnification m_2 will satisfy the conditions,

(1) $V_2 > V_1$ and $m_2 > m_1$
(2) $V_2 > V_1$ and $m_1 > m_2$
(3) $V_2 < V_1$ and $m_2 > m_1$
(4) $V_2 < V_1$ and $m_1 > m_2$
(5) $V_2 < V_1$ and $m_1 = m_2$

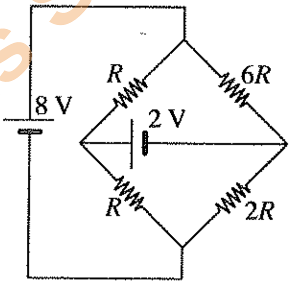
39. Two point charges of $+q$ each, are held at vertices B and C of an equilateral triangle ABC of side length a, and another point charge of $+q$ is held at the point P as shown in the figure. A zero resultant force will act on a positive unit charge placed at point A when the distance AP is equal to



(1) $\sqrt{2}a$ (2) $\frac{a}{2}$ (3) $\frac{a}{\sqrt{(\sqrt{3})}}$

(4) $\frac{a}{4}$ (5) a

40. In the circuit shown, the two cells have negligible internal resistances. In the circuit,



(1) a current of $\frac{3}{2R}$ passes through the 2 V cell.

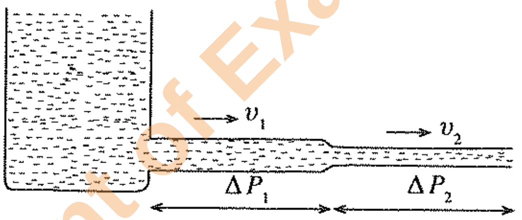
(2) a current of $\frac{6}{R}$ passes through the 2 V cell.

(3) a current of $\frac{10}{R}$ passes through the 2 V cell.

(4) a current of $\frac{3}{R}$ passes through the 2 V cell.

(5) a current does not pass through the 2 V cell.

41. Two narrow tubes of equal lengths but different radii of cross-section are connected end to end, and water is allowed to flow through it as shown in the figure.

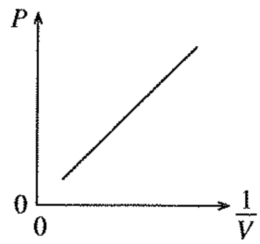


If v_1 and v_2 are the average velocities with which water flows through cross-sections of the tubes, and ΔP_1 and ΔP_2 are the pressure differences built up across the tubes as shown, then the ratio, $\frac{\Delta P_1}{\Delta P_2}$ is equal to

(1) $\left(\frac{v_1}{v_2}\right)^{\frac{1}{4}}$ (2) $\frac{v_1}{v_2}$ (3) $\left(\frac{v_1}{v_2}\right)^2$ (4) $\left(\frac{v_1}{v_2}\right)^3$ (5) $\left(\frac{v_1}{v_2}\right)^4$

42. A student performed an experiment to verify the Boyle's Law using a constant mass m_0 of an ideal gas at the room temperature of 27 °C and obtained the graph given in the figure. Here P is the pressure and V is the volume of the gas.

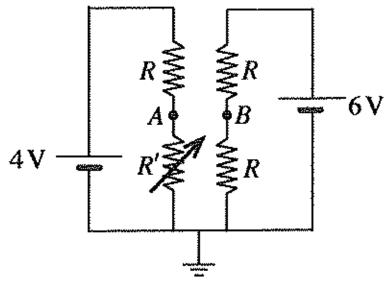
He then removed a certain amount of gas from the volume V and repeated the experiment at a temperature 100 °C above the room temperature. If the new graph he obtained has the same gradient as the graph shown in the figure, the mass of the gas that he had removed is

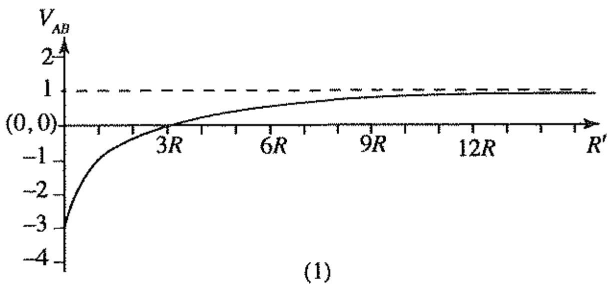


(1) $\frac{27}{100}m_0$ (2) $\frac{73}{100}m_0$ (3) $\frac{1}{4}m_0$ (4) $\frac{1}{2}m_0$ (5) $\frac{3}{4}m_0$

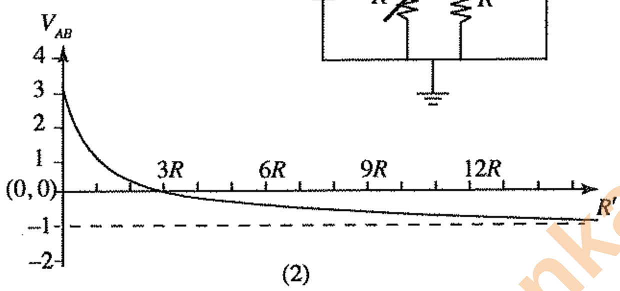
[See page eight

43. In the circuit shown, both cells have negligible internal resistances. R' is the value of a variable resistor. Variation of the voltage V_{AB} (= $V_A - V_B$) across the points A and B with R' is best represented by

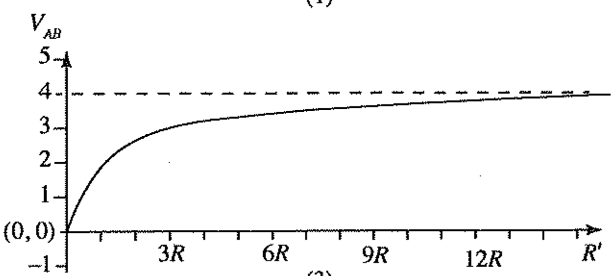




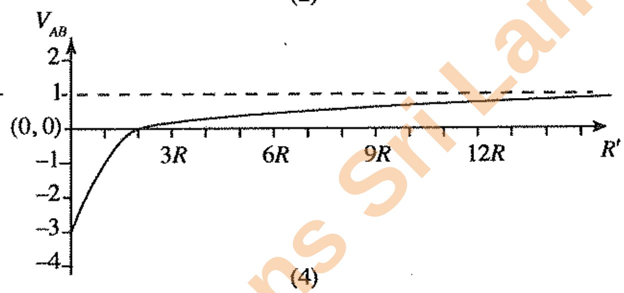
(1)



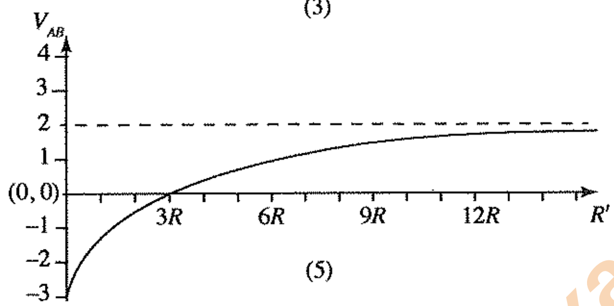
(2)



(3)

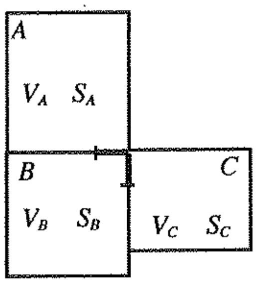


(4)

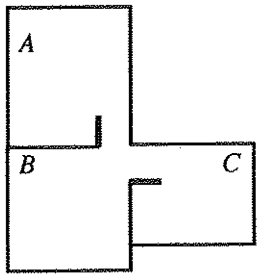


(5)

44. Absolute humidities of air inside three closed rooms A, B and C of volumes V_A, V_B and V_C at atmospheric pressure are S_A, S_B and S_C respectively. [See figure (a)]. The dew point of air in room A is T_0. When the doors are opened as shown in figure (b) and the air in three rooms are allowed to mix, the common dew point of the three rooms will remain at T_0 if



(a)



(b)

(1) $S_A = \dfrac{V_B S_B + V_C S_C}{V_B + V_C}$

(2) $S_A = \dfrac{S_B + S_C}{2}$

(3) $V_A S_A = V_B S_B + V_C S_C$

(4) $\dfrac{S_A}{V_A} = \dfrac{S_B}{V_B} + \dfrac{S_C}{V_C}$

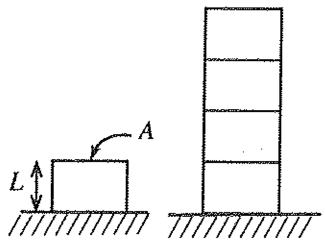
(5) $S_A = \sqrt{S_B S_C}$

45. A 2 μF capacitor and a 1 μF capacitor are connected in series and charged by a battery. Then the stored energies of the capacitors are E_1 and E_2 respectively. When they are disconnected, allowed to discharge, and charged again **separately** using the same battery, the stored energies of the two capacitors are E_3 and E_4 respectively. Then

(1) $E_3 > E_1 > E_4 > E_2$
(2) $E_1 > E_2 > E_3 > E_4$
(3) $E_3 > E_1 > E_2 > E_4$
(4) $E_1 > E_3 > E_4 > E_2$
(5) $E_3 > E_4 > E_2 > E_1$

[See page nine

46. The height of a rectangular heavy metal block of mass M, area of cross-section A, and made of a material of Young's modulus Y, when placed on a horizontal surface as shown in figure (a) is L. If four blocks identical to the above mentioned block are stacked together as shown in figure (b), the overall height of the four blocks will be

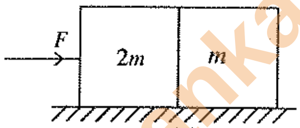


(a) (b)

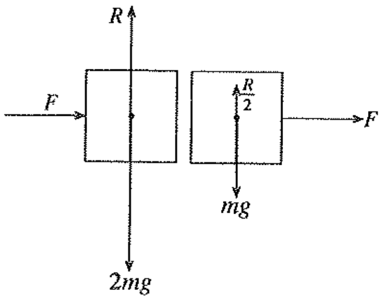
(1) $L\left(4-\dfrac{2Mg}{YA}\right)$ (2) $L\left(4-\dfrac{8Mg}{YA}\right)$ (3) $L\left(4-\dfrac{7Mg}{YA}\right)$

(4) $L\left(4-\dfrac{6Mg}{YA}\right)$ (5) $L\left(4-\dfrac{4Mg}{YA}\right)$

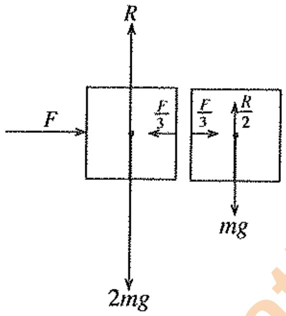
47. Two blocks of mass $2m$ and m are placed in contact on a smooth surface as shown in the figure (a). If an external horizontal force F is applied on the block of mass $2m$, which of the following figures shows the forces acting on the two blocks correctly?



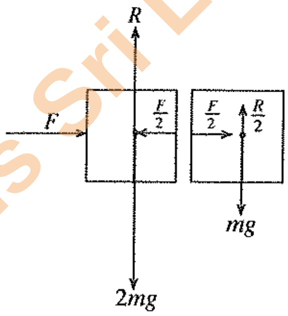
(a)



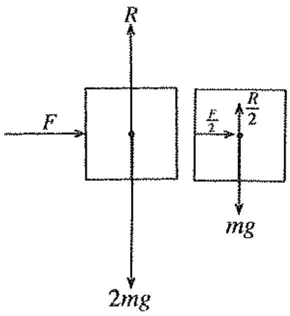
(1)



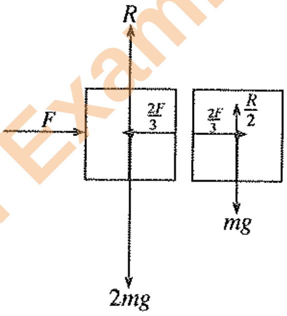
(2)



(3)

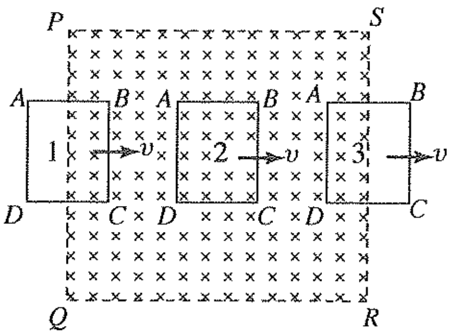


(4)



(5)

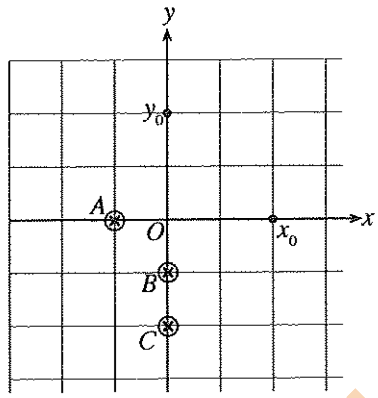
48. As shown in the figure, a rectangular wire loop $ABCD$ is inserted perpendicular to a uniform magnetic field confined to a region $PQRS$ from position 1 and taken across the field with a constant velocity v. It passes through position 2 and finally taken out of the magnetic field at position 3 with the same velocity. Which of the following statements is **not true**?



(1) When the loop passes through position 1, a constant e.m.f. will be induced only across section BC of the wire loop.
(2) As the loop passes through position 2, constant e.m.f.s will be induced across AD and BC, and they are equal and opposite to each other.
(3) At position 3, a constant e.m.f. will be induced only across AD.
(4) At position 2, the resultant force on the loop due to magnetic field is zero.
(5) The directions of the forces due to magnetic field on the loop at positions 1 and 3 are opposite to each other.

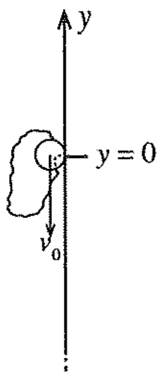
[See page ten

49. Three thin long and straight wires carrying equal currents I are held in fixed positions A, B and C perpendicular to the plane of the paper as shown in the figure, where $OA = 1$ m, $OB = 1$ m and $OC = 2$ m. Two other thin, long and straight wires are also held perpendicular to the plane of the paper, at points x_0 and y_0 where $x_0 = 2$ m and $y_0 = 2$ m. Which of the following currents set up in the wires at x_0 and y_0 will produce a resultant magnetic field of magnitude $\frac{\mu_0 I}{2\pi}$ in positive y-direction at the point O.



	Current to be set up in the wire at x_0	Current to be set up in the wire at y_0
(1)	$3I$ ⊙	$4I$ ⊗
(2)	$4I$ ⊙	$6I$ ⊙
(3)	$4I$ ⊗	$3I$ ⊗
(4)	$4I$ ⊗	$4I$ ⊙
(5)	$6I$ ⊙	$4I$ ⊙

50. A particle of mass m is attached to one end of a light elastic string of force constant k and unstreatched length of l_0. The other end of the string is fixed onto a vertical frictionless wall at $y = 0$ as shown in the figure. The particle is then projected vertically downwards from the position $y = 0$ with a velocity v_0, $\left(v_0 < \sqrt{2gl_0}\right)$. Neglect the air resistance.



After passing through its lowest point in the path, the particle will again come to rest momentarily at a point whose y coordinate is

(1) $-\frac{\left[m(v_0^2 + 2gl_0) - kl_0^2\right]}{2gm}$

(2) $-\frac{(v_0^2 + 2gl_0)}{2g}$

(3) $\frac{v_0^2 + 2gl_0}{2g}$

(4) $\frac{mv_0^2 + kl_0^2}{gm}$

(5) $\frac{v_0^2}{2g}$

* * *

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இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2016 අගෝස්තු
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2016 ஓகஸ்ற்
General Certificate of Education (Adv. Level) Examination, August 2016

භෞතික විද්‍යාව II பௌதிகவியல் II **Physics II**	**01 E II**	පැය තුනයි மூன்று மணித்தியாலம் *Three hours*

Index No. : ....

Important :

- ✻ *This question paper consists of* **13** *pages.*
- ✻ *This question paper comprises of* ***two*** *parts, Part* **A** *and Part* **B.** *The time allotted for* ***both parts*** *is* ***three hours.***
- ✻ *Use of calculators is* ***not*** *allowed.*

PART A — Structured Essay :
(pages 2 - 7)

Answer ***all*** *the questions on this paper itself. Write your answers in the space provided for each question. Note that the space provided is sufficient for your answers and that extensive answers are not expected.*

PART B — Essay :
(pages 8 - 13)

This part contains ***six*** *questions, of which,* ***four*** *are to be answered. Use the papers supplied for this purpose.*

- ✻ *At the end of the time allotted for this paper, tie the* ***two parts together*** *so that* ***Part A*** *is* ***on top*** *of* ***Part B*** *before handing them over to the Supervisor.*
- ✻ *You are permitted to remove* ***only Part B*** *of the question paper from the Examination Hall.*

For Examiners' Use Only

For the second paper

Part	Question Nos.	Marks Awarded
A	1	
	2	
	3	
	4	
B	5	
	6	
	7	
	8	
	9 (A)	
	9 (B)	
	10 (A)	
	10 (B)	
Total		

Final Marks

In numbers	
In words	

Code Numbers

Marking Examiner 1	
Marking Examiner 2	
Marks checked by	
Supervised by	

[see page two

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PART A — Structured Essay

*Answer **all four** questions on this paper **itself**.*

(Acceleration due to gravity, $g = 10\,\mathrm{N\,kg^{-1}}$)

1. When certain objects are packed in containers they do not occupy the entire volume of the container. This occurs due to the shape of the objects, and under such situations a fraction of the container volume is always empty and filled with air. Consider a container in the form of a cubical box of side length $8r$, which is fully packed with identical solid spheres of **radius** r in a regular manner as shown in figure (1). This is called **regular packing**.

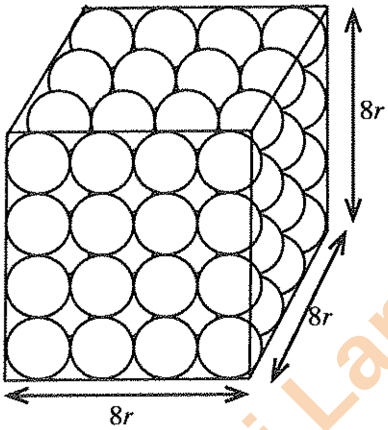

Figure (1)

(*a*) Find the number of spheres packed in the container.

....

(*b*) Obtain an expression for the total material volume of all spheres packed in the container in terms of r and π.

....

(*c*) When the container is completely filled with spheres, the ratio,

$$\frac{\text{Total material volume of the spheres in the container}}{\text{Volume of the fully packed container}}$$, is called the **packing fraction** (f_p)

of the spheres and the volume of the fully packed container is called the **packed volume**.

Find the packing fraction, f_p, for the above regular packing.

....

....

....

(*d*) If m is the total mass of the spheres in the container, derive an expression for the ratio:

$$\frac{\text{Total mass of the spheres}}{\text{Volume of the fully packed container}}$$, in terms of m and r.

This ratio is called the **bulk density** (d_B) of the spheres.

....

....

(*e*) Write down an expression for the density (d_M) of the material of the spheres in terms of m, r and π.

....

....

(*f*) A student has decided to find the parameters f_p, d_B and d_M for green gram using an experimental method. In this case green gram is packed in a random manner and it is called **random packing**. See figure (2). The definitions mentioned in part (*c*), (*d*) and (*e*) for f_p, d_B and d_M are valid for random packing of items of any shape too.

First he inserted dry green gram into a measuring cylinder and obtained a packed volume of 50 cm^3 of green gram as shown in figure (2).

Then he measured the mass of the packed volume 50 cm^3 sample of green gram and it was found to be 3.8×10^{-2} kg.

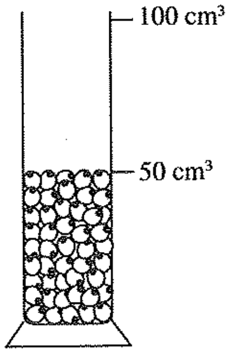

Figure (2)

[see page three

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Subsequently he introduced the green gram sample into a measuring cylinder containing 50 cm^3 of water and found that the water level raised to 82 cm^3 mark. See figure (3).

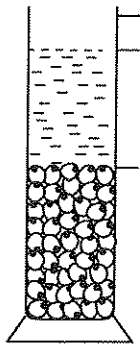

Figure (3)

(i) What is the material volume of green gram?

....

(ii) Calculate the packing fraction (f_p) of green gram.

....

....

(iii) Calculate the bulk density (d_B) of green gram in kg m^{-3}.

....

....

(iv) Calculate the density (d_M) of material of green gram in kg m^{-3}.

....

....

(*g*) A polythene bag is to be designed to pack 1 kg of green gram. Calculate the minimum volume of the bag needed.

....

....

2. You are asked to determine the dew point of air in the laboratory experimentally, and determine its relative humidity.

(*a*) Write down an expression for the relative humidity (RH) in terms of saturated vapour pressures.

RH = ...

....

(*b*) In addition to a polished calorimeter with a lid and a stirrer, what other items would you require to carry out this experiment?

....

....

(*c*) Write down **two** factors that need to pay attention **before starting** the experiment in order to obtain a final result with better accuracy, and state experimental precautions that you would take to minimize them.

	Factors	**Experimental precautions**
(1)		
(2)		

(*d*) Small pieces of ice are used for this experiment. Give reasons for this.

....

....

[see page four

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(*e*) What practical difficulties would you face if several pieces of ice are added to water at a time?

....

....

(*f*) Exactly at what instants do you take the readings in this experiment?

....

....

(*g*) What is the reason for using the calorimeter with a lid in this experiment?

....

(*h*) What is the other reading that you should take in this experiment?

....

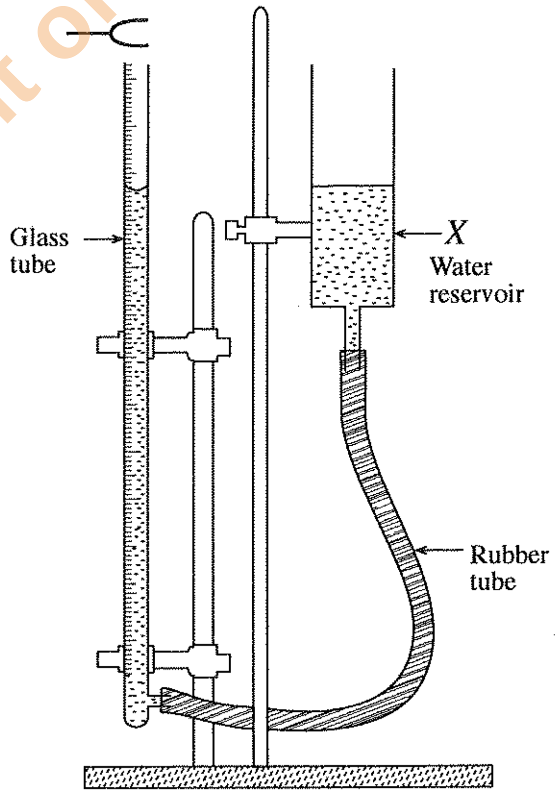
(*i*) When the temperature of a certain laboratory was 28 °C, its dew point was found to be 24 °C. Using the following table, determine the relative humidity of the laboratory.

Temperature (°C)	20	22	24	26	28	30	32
Saturated water vapour pressure (mmHg)	17.53	19.83	22.38	25.20	28.35	31.82	35.66

....

....

3. Figure shows an alternative apparatus to find the speed of sound in air using a resonance tube with one end closed. The principle of this apparatus is similar to the principle of the apparatus normally used in the school laboratory. The resonance tube in this apparatus is a glass tube with a calibrated scale. The water level in the resonance tube is raised and lowered by raising and lowering of a water reservoir X which is connected to the resonance tube with a flexible rubber tube.



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(*a*) What type of wave is formed inside the tube at resonance?

....

(*b*) You are given a tuning fork of known frequency f and asked to obtain the resonant lengths l_0 and l_1 corresponding to the fundamental note and the first overtone respectively.

(i) Draw the wave patterns for the two modes of vibrations, and mark the lengths l_0, l_1, end correction e, Nodes (N) and Anti-nodes (AN).
(You are expected to draw the tube for the first overtone).

Fundamental note: 

First overtone:

(ii) (1) If λ is the wavelength corresponding to the fundamental note, write down an expression for λ in terms of, l_0 and e.

....

(2) Write down a similar expression for the wave length corresponding to the first overtone.

....

(3) If v is the speed of sound in air, derive an expression for v in terms of the known and measured quantities.

....

....

....

(*c*) Before taking the measurement for l_0, the water level in the resonance tube has to be raised upto the top. Explain the reason for this.

....

(*d*) Write down **two** major differences in the experimental procedure when using the apparatus given in the question compared to the method adapted when using apparatus generally available in school laboratory.

(1) ....

(2) ....

(*e*) At room temperature (28 °C), when a 512 Hz tuning fork is used, the corresponding lengths of the resonance for fundamental note and the first overtone are found to be 15.5 cm and 50.5 cm respectively.
Calculate the speed of sound in air at room temperature.

....

....

....

[see page six

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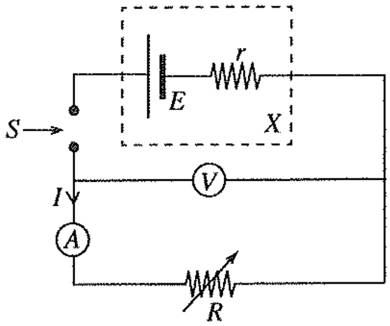
4. The given circuit can be used in a school laboratory to experimentally determine the e.m.f. (E), and the internal resistance (r) of a dry cell X using a graphical method.

The experimental procedure consists of measuring the potential difference V across the terminals of the cell for different values of I using a voltmeter with very high internal resistance.



(a) Write down an expression for V in terms of I, E, and r.

....

(b) (i) Name the variable resistor that is available in the school laboratory, which could be used for this experiment.

....

(ii) The key S should be used properly to obtain expected results from this experiment.

(1) What is the type of key that is most suitable to be used as S?

....

(2) What is the experimental procedure you adapt when operating the key?

....

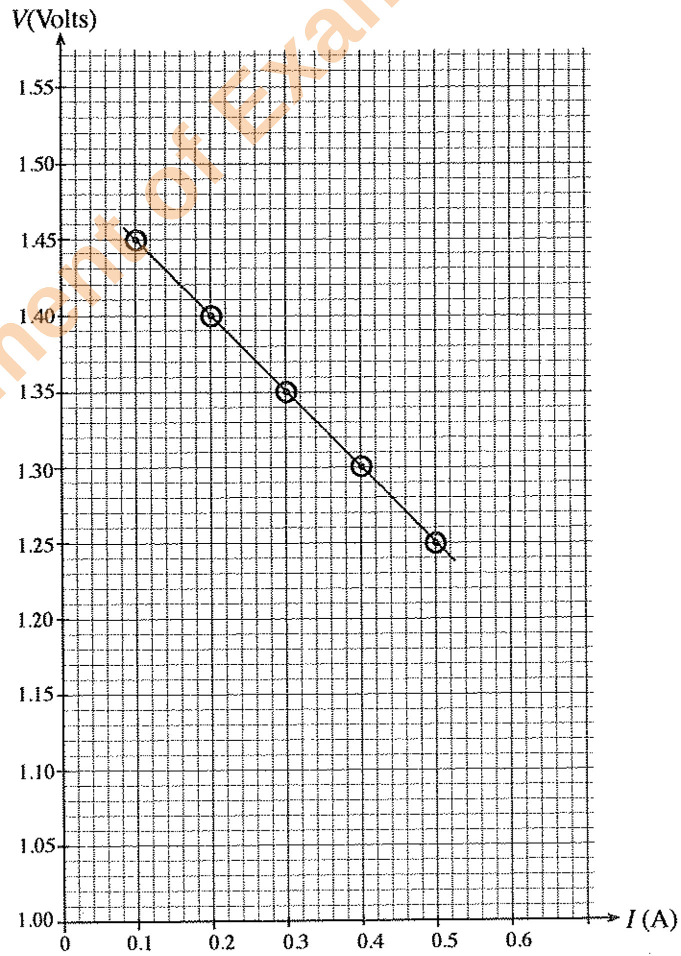
....

(iii) How do you confirm experimentally that the cell has not run down while carrying out the experiment?

....

....

(c) A graph of V against I plotted using a set of data obtained from such an experiment is shown below.



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(i) Use the graph to find the following.

(1) Internal resistance, r of the cell.

....

....

(2) E.m.f. E of the cell.

....

(ii) Use the values obtained under (c)(i) and the expression obtained under (a) to deduce the current (I_{SC}) through the cell if it is short circuited.

....

(d) A supply voltage in the range, 8.6 V – 9.0 V will have to be applied to operate a certain electronic item properly. The resistance across the supply voltage terminals of the electronic item is 30 Ω. Suppose you have a choice of selecting a single dry cell battery having E = 9 V and r = 10 Ω **or** a combination of six dry cell batteries each having E = 1.5 V and r = 0.2 Ω and connected in series, for the operation of the above electronic item. Using the data given in this part, explain how you would select a proper battery.

....

....

....

* *

[see page eight

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கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2016 ஓகஸ்ட்
General Certificate of Education (Adv. Level) Examination, August 2016

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பௌதிகவியல் II
Physics II

01 E II

PART B — Essay

Answer ***four*** *questions only.*

(Acceleration due to gravity $g = 10\,\text{N kg}^{-1}$)

5. A particle of mass m_1 is fixed to the rim of a horizontal ring of radius r and negligible mass as shown in figure (1). POQ is a vertical axis passing through the centre O of the ring.

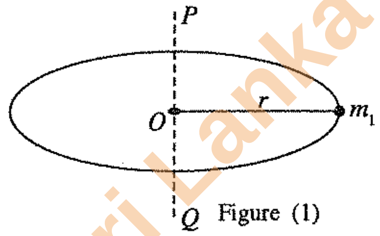

Figure (1)

(*a*) (i) Write down an expression for the moment of inertia I_1 of the particle about the vertical axis POQ in terms of m_1 and r.

(ii) Another particle of mass m_2 is now fixed to the rim of the ring which is diametrically opposite to m_1, and the system is rotated about the axis POQ with a constant angular speed ω. If I_2 is the moment of inertia of mass m_2 about the axis POQ, write down an expression for the total rotational kinetic energy (E) of the system.

(iii) If I_0 represents the total moment of inertia of the above system in (*a*) (ii) about the axis POQ, using the expression obtained in (*a*)(ii) show that $I_0 = I_1 + I_2$.

(*b*) Instead of m_1 and m_2, 10 identical particles, each of mass m, are now fixed to the rim of the ring with equal spacing. If I is the moment of inertia of a particle about the vertical axis POQ, write down an expression for total moment of inertia (I_T) of the system about the vertical axis POQ.

(*c*) Now, the ring described in (*b*) above is fixed onto an axle of negligible moment of inertia and coinciding with the vertical axis POQ using symmetrically fixed spokes of negligible mass as shown in the figure (2). The system is then started rotating from rest at time $t = 0$ in a horizontal plane about the axis POQ with a constant angular acceleration α and reached a constant angular speed ω.

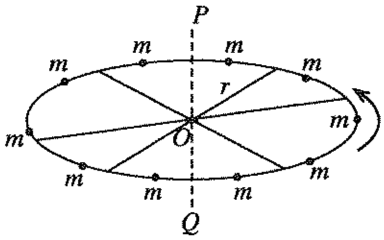

Figure (2)

(i) (1) Obtain an expression for the time t taken by the system to reach the constant angular speed ω.

(2) How many revolutions have been made by the system when it reaches the constant angular speed ω?

(ii) Write down an expression for the centripetal force (F) acting on one particle when it is rotating about the axis POQ with a constant angular speed ω.

(*d*) The structure of the merry-go-round shown in figure (3) which is at rest is similar to the structure of the system described in (c) above. However, instead of fixed masses m, the system has 10 chairs occupied by riders and hung by chains of negligible mass. The moment of inertia of the merry-go-round, **without riders and chairs**, about the axis POQ is 32 000 kg m^2.

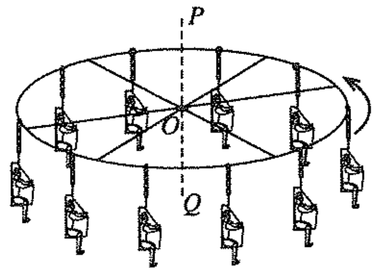

Figure (3)

Consider a situation where the merry-go-round is rotating about the axis POQ with a constant angular speed of 12 revolutions per minute with all the chairs being occupied by riders. When the merry-go-round rotates, all the chains are inclined to the vertical by an angle θ, and figure (4) shows the situation with respect to one rider.

Use $\pi = 3$ for relevant calculations.

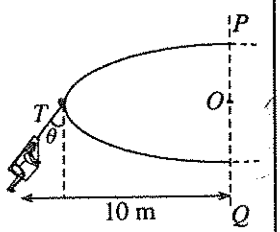

Figure (4)

(i) If the riders are of mass 70 kg each and the chairs are of 20 kg each, calculate the total moment of inertia of the system about the axis POQ. When calculating the moment of inertia assume that the total mass of the rider and his chair **is concentrated** at a horizontal distance of 10 m from the axis POQ.

(ii) Calculate the value of θ.

(iii) What is the total rotational kinetic energy of the system?

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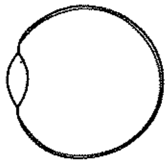
6. The effective focal length of the cornea and the eye-lens can be considered as the focal length of an eye. The muscles controlling the curvature of the lens permit the eye to focus on the retina light from objects at different distances from the eye. The figure shows a simplified diagram of the eye with an eye-lens of effective focal length. When the eye muscles are relaxed the focal length of a healthy eye of a child is about 2.5 cm. The near point of his eye is at a distance of 25 cm.



(Copy the diagram given in the figure and use it when drawing ray diagrams.)

(*a*) Draw a ray diagram for the situation where light from a far away object is focused onto the retina of the eye of the child with healthy eye when his eye muscles are relaxed. What is the distance between the eye-lens and the retina?

(*b*) Draw a ray diagram for a situation where a point source of light is placed at the near point, is clearly seen by the child with healthy eye. Calculate the focal length of the eye at this instant.

(*c*) Another child has the focal length equal to that of the healthy child when the eye muscles are relaxed and also has the focal length calculated for the situation in (*b*). However, the position of his retina is located 0.2 cm behind the position of the retina of the healthy child.

(i) Using the image produced by a point source of light as mentioned in (b) above, indicate his near point and far point by drawing two separate ray diagrams. Calculate the distances from the eye-lens to the near point and to the far point of this child.

(ii) Sketch a ray diagram illustrating as to how the required correction can be done using a suitable lens. Calculate the focal length of the corrective lens needed.

(*d*) When a person becomes older the ability to change the focal length of eyes gets weaker and the distance to the near point of the eye increases. If the child mentioned in part (*c*) above would face such a situation, what is the type of additional corrective lens that the child should wear (convergent/divergent)? Give reasons for your answer.

7. Write down Poiseuille's equation for the rate of flow, Q, of a liquid through a horizontal cylindrical narrow tube under a pressure difference of ΔP. Identify all the other symbols you used.

Under the above condition, the resistance exerted by the tube against the rate of flow of the liquid, Q, can be defined as the flow resistance $R = \dfrac{\Delta P}{Q}$.

(*a*) What physical quantities associated with the tube and the liquid determine the flow resistance R?

(*b*) When a liquid flows under pressure differences of ΔP_1, ΔP_2 and ΔP_3, through three horizontal narrow tubes connected in series as shown in figure (1), the flow resistances exerted by the tubes are R_1, R_2 and R_3 respectively. Using the definition given above for R show that the flow resistance, R_0, of the system can be written as $R_0 = R_1 + R_2 + R_3$. (Neglect edge effects.)

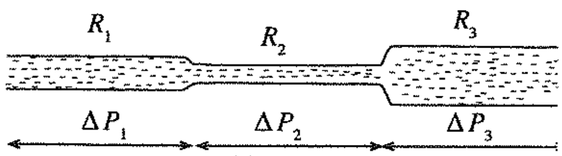


Figure (1)

(*c*) When a liquid flows through two horizontal narrow tubes connected in parallel under a common pressure difference ΔP as shown in figure (2), the flow resistances exerted by the tubes are R_1 and R_2. Show that the flow resistance R_0 of the system can be written as $\dfrac{1}{R_0} = \dfrac{1}{R_1} + \dfrac{1}{R_2}$. (Neglect end effects.)

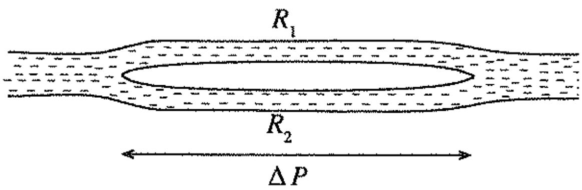


Figure (2)

(*d*) Figure (3) shows a set of horizontal narrow tubes A, B, C, D and E connected between the point X and a common reservoir Y so that a liquid can flow from X to Y. The pressures at X and Y are maintained at constant values. The flow resistance of each tube is labelled in the diagram in units of mmHg s/cm^3. Tube B is divided into two tubes C and D of equal flow resistances. This simplified model may also be used to illustrate the blood flow through arteries and veins.

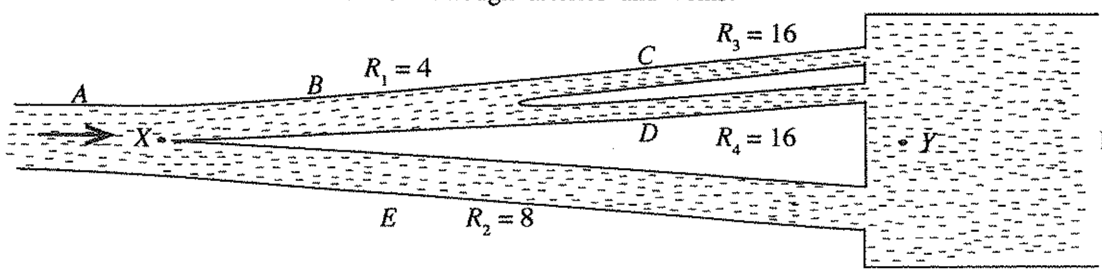


Figure (3)

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Give the answers to parts (i), (ii) and (iii) in terms of the given units. (Take $\pi = 3$)

(i) (1) Calculate the flow resistance, due to the system of tubes B, C and D between the points X and Y.

(2) Calculate the flow resistance, due to the system of tubes, B, C, D and E between the points X and Y.

(ii) If the flow rate of the liquid across X is 6 cm^3/s, calculate the pressure difference between X and Y.

(iii) Using the above results, find the flow rate of the liquid through tube E.

(iv) If the length of tube E is 2 cm find the internal radius of tube E. The viscosity of the liquid is 4.0×10^{-3} Pa s [Take 1 mmHg = 133 Pa]

(*e*) If the temperature of one of the tubes in the system given in part (*d*) is reduced, explain what would happen to the flow rate of the liquid in that tube. Neglect the changes in radius and length of the tube.

8. Read the following passage and answer the questions.

Induction heating technology is of choice in many industrial, domestic and medical applications due to its advantages such as less heating time, localized heating, direct heating and efficient energy consumption. The operating principle of induction heating is based on the law of the electromagnetic induction discovered by Michael Faraday in 1831. The two major components in an induction heating system are a coil of wire (often a copper coil) producing a time varying magnetic field upon receiving a high frequency alternating current, and an electrically conducting material that generates heat. The magnetic field also changes its direction as the direction of the alternating current changes. When a conducting material is exposed to such a time-varying magnetic field, current loops called eddy currents are induced in the conducting material. As the magnetic field changes its direction rapidly the eddy currents also change their directions rapidly. The eddy currents always form closed loops inside conducting materials in planes perpendicular to the varying magnetic field. Eddy currents, generate Joule heat (I^2R type heat) due to the existence of resistance of the material.

When the magnetic field created is stronger or when the electrical conductivity is higher or when the rate of change of magnetic field is larger, the eddy currents that are developed become larger. The eddy currents which are generated by high frequency alternating current in the coil will exist only within a limited thickness near the surface of the material due to what is called skin effect.

The skin effect is the tendency of any high frequency electric current to distribute itself in a conductor with the current density being largest near the surface of the conductor and decreasing very rapidly with the depth of the conductor. This thickness across which eddy currents are distributed becomes even smaller due the mutual attraction between the alternating current in the coil and the eddy current loops. This is called the proximity effect. In addition to the Joule heating, an additional heat is also produced within the material due to a phenomenon called hysteresis effect. It occurs only in ferromagnetic materials such as some stainless steel, cast iron, nickel, etc. In response to the varying magnetic field produced by the alternating current, the magnetic domains in these materials repeatedly change their orientations. The energy required to turn them around finally is converted to heat. The rate at which the heat is generated due to hysteresis effect increases with the frequency of the varying magnetic field. Commercially available induction heating systems operate at frequencies approximately from 60 Hz to about 1 MHz and deliver power in the range from a few watts to several Megawatts.

The cookers that are available in the market as induction cookers operate on this principle. In an induction cooker, a coil of copper wire is mounted just under the surface of the cooker top where the cooking pot is placed, without touching it, and an alternating electric current is sent through the coil. The entire bottom of the cooking pot itself acts as the conducting material that generates the heat. The varying magnetic field produced by the coil enters the bottom of the cooking pot creating eddy currents and hysteresis losses, generating heat. In order to make use of both effects for heat generation, the cooking pots or the bottoms of the cooking pots are made of ferromagnetic materials such as some stainless steel or cast iron.

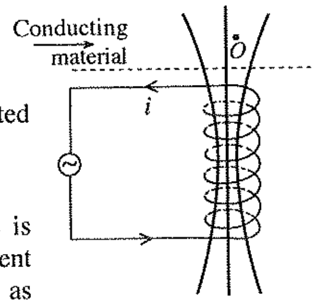
(*a*) State Faraday's Law of electromagnetic induction in words.

(*b*) Name **two** fields of application where induction heating is used.

(*c*) Write down the **two** heating processes involved in the induction heating.

(*d*) Write down **three** factors which give rise to larger eddy currents.

(*e*) Write down the **two** effects which limit the eddy currents to be within a limited thickness near the surface of the material.



(*f*) Copy the given diagram and answer the following questions.

The direction of the alternating current in a coil at a certain instant of time is shown in the figure. Consider a situation where the magnitude of this current is increasing with time. A conducting material is placed just above the coil as shown in the figure.

(i) Show the direction of the magnetic field created in this situation by drawing an arrow on one field line.

(ii) Draw one loop of eddy current in the material near the position O and show the direction of the eddy current when the alternating current is increasing.

(iii) Use Lenz's law to explain how you determined the direction of the eddy current loop that you have drawn in (ii) above.

(*g*) Explain how the increase of the frequency of alternating current, increases the rate of heating in the material.

(*h*) Consider a situation where a time-varying magnetic field enters a disk of radius R, thickness b and resistivity ρ. If the flux density B of the applied magnetic field varies sinusoidally as $B = B_0 \sin \omega t$ where B_0 is the amplitude of the flux density of the magnetic field, ω is the angular frequency and t is the time, then based on a very simplified model the average power P generated by the eddy currents in the disk can be given by $P = k\, B_0^2\, \omega^2$ where $k = \dfrac{\pi R^4 b}{16\rho}$. If $k = 0.5\ \mathrm{m^4\ \Omega^{-1}}$, $\omega = 6000\ \mathrm{rad\ s^{-1}}$ and $B_0 = 7.5 \times 10^{-3}$ T, calculate the average power generated in the disk.

(*i*) In transformers, the core is heated up due to eddy currents and it contributes to energy loss in the form of heat. How is this energy loss minimised in transformers?

9. Answer either part (A) or part (B) only.

(A) (*a*) Write down an expression for the energy (W) dissipated in a resistor of resistance R when a current of magnitude I is passed through it for a period of time t.

(*b*) An electrical fuse is a small element consisting of a thin metal wire. Electrical fuses are connected in series with electrical/electronic circuits to avoid damages caused to them due to the passage of currents larger than the recommended current for the circuits (due to over-load currents and short circuits). When the current through the fuse in a certain circuit becomes larger than the recommended current in the circuit, the fuse burns (melts) and disconnects the circuit from the power source. The electrical fuses are selected so that their ratings are equal to the recommended currents in the circuits.

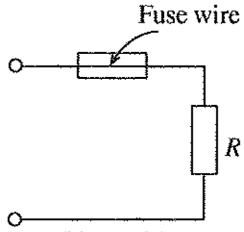


Figure (1)

(i) Figure (1) shows how a fuse is connected to a circuit of load resistance R. Current in a certain fuse is rated as 5 A. If the length of the fuse wire is 3 cm, and its radius is 0.1 mm (area of cross-section $\sim 3 \times 10^{-8}\ \mathrm{m^2}$) and the resistivity of the material of the wire at 25 °C is $1.7 \times 10^{-8}\ \Omega$ m, calculate the resistance of the fuse wire at room temperature of 25 °C.

(ii) When the fuse is operated at the rating mentioned in (i), at steady state, the entire heat generated by the fuse wire is dissipated to the surrounding without burning the fuse. Calculate the power dissipated by 5 A fuse in that manner. Take the average value of the resistance of the fuse wire over the temperature range is equal to five times the resistance calculated under (*b*)(i).

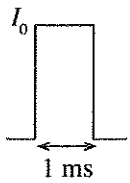


Figure (2)

(iii) A test performed by manufacturers of electrical fuses is to determine the amplitude of a current pulse needed to melt (burn) the fuse wire approximately in one millisecond. Considering the rectangular current pulse of one millisecond duration shown in the figure (2), calculate the peak current I_0 of the pulse needed to melt the fuse wire given in (*b*)(i). Assume that the heat dissipation to the surroundings under this condition is negligible. Take the mass of the fuse wire given in (*b*)(i) as 7.5×10^{-6} kg, and the average value of resistance of the fuse wire as five times the resistance calculated under (*b*)(i). Specific heat capacity of the material of the fuse wire is 390 J kg^{-1} °C^{-1}. Melting point of the material of the fuse wire is 1 075 °C.

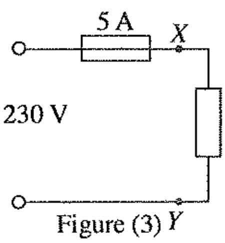


Figure (3)

(iv) Consider a situation in which a load circuit with an applied voltage of 230 V as shown in the figure (3) is short circuited at XY. Calculate the current through a 5 A fuse under this situation. Using the results obtained in (*b*)(iii), show that the fuse will melt before 1 millisecond. (Assume that the current produced is a rectangular current pulse.)

(v) A rectangular narrow current pulse of 500 A occurring for a duration of 1 μs passes through a 5 A fuse. In this situation, will the fuse get burnt? Justify your answer using an appropriate calculation.

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(B) Figure (1) shows the circuit symbol of an operational amplifier having open loop voltage gain A.

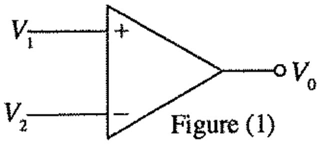

Figure (1)

(*a*) Write down the expression for the output voltage V_0 in terms of V_1, V_2, and A.

(*b*) If the positive and negative output saturation voltages of the operational amplifier are ± 15 V and $A = 10^5$, calculate the minimum input voltage difference which will drive its output into saturation.

(*c*) (i) Draw the output voltage waveform when the given triangular voltage signal of peak amplitude 5 V is applied to the + input of the circuit as shown in figure (2), and label its peak voltage values.

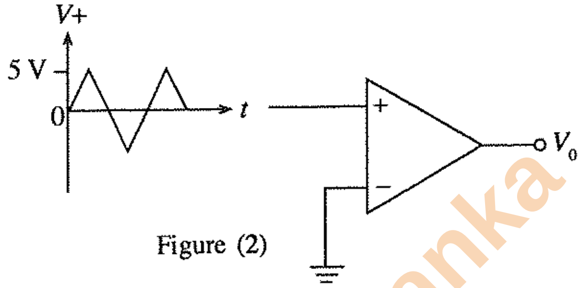

Figure (2)

(ii) The circuit in figure (2) is now modified as shown in figure (3). When S_1 is closed and S_2 is open the circuit will produce the output waveform shown in the figure (3) for the input triangular signal. Considering the actions of circuit elements in figure (3), explain the reasons for differences, if any, between the output voltage waveform shown in figure (3) and the waveform drawn by you in (*c*)(i). What is the peak voltage of the output in figure (3)?

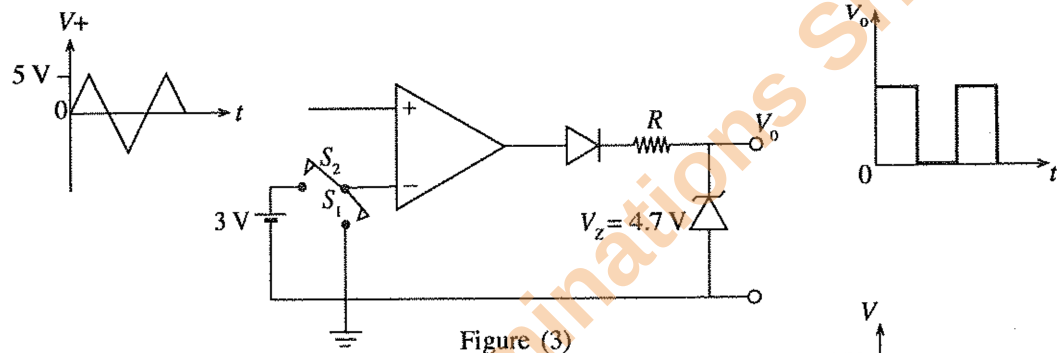

Figure (3)

(iii) Now a voltage of +3 V is applied to the − input of the operational amplifier in figure (3) by opening S_1 and closing S_2. When a hypothetical voltage waveform shown in figure (4) is applied to the + input of the operational amplifier, draw the output waveform expected from the circuit and label the magnitude of the output voltage.

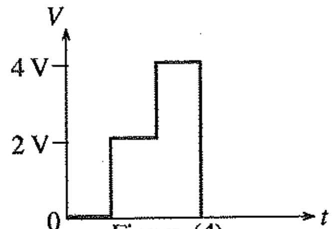

Figure (4)

(*d*) A certain blood cell counting system operates as follows. The blood is diluted by a known proportion in a proper type of solution, and allowed to flow through a small aperture X of the order of 50 μm diameter placed in between two electrodes S and T as shown in the figure (5). Blood cell counting is based on the fact that the electrical resistivity of blood cells is higher than the electrical resistivity of the solution.

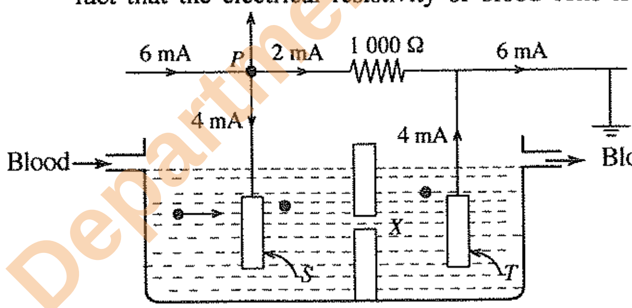

Figure (5)

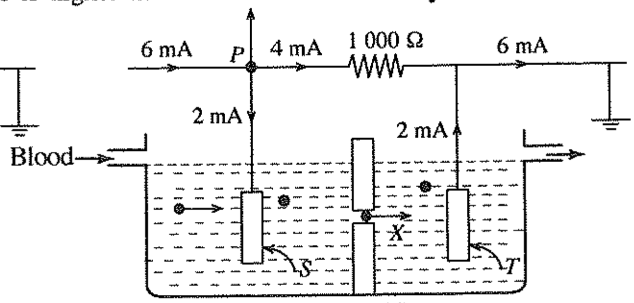

Figure (6)

A constant current of 6 mA is passed through the system as shown in figures (5) and (6). Currents through 1000 Ω resistor and the electrodes when the **solution** passes through the aperture X is shown in figure (5). Figure (6) shows the currents through 1 000 Ω resistor and the electrodes when a **blood cell** is going through the aperture X. The point P of the circuits shown in figures (5) and (6) is connected to + terminal of the operational amplifier in the circuit shown in figure (3) with S_1 open and S_2 closed. The output V_0 is connected to a pulse counter. (Not shown in the figure.)

(i) What are the voltages at point P in figures (5) and (6)?

(ii) If the situation in figure (5) occurs before (6), draw the voltage waveform at P for such situations.

(iii) Draw the output voltage waveform of the circuit in figure (3) relevant to (ii) above.

(iv) What does the counter output indicate if a diluted blood stream is allowed to flow through the aperture X?

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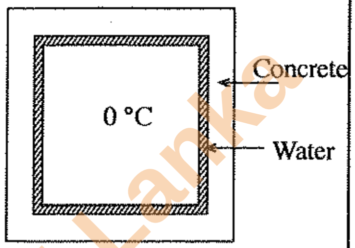
10. Answer either part (A) or part (B) only.

(A) (*a*) (i) Briefly explain how heat is absorbed when the physical state of a material is changed from solid state to the liquid state.

(ii) 10 mega Joules of excess thermal energy produced by a certain thermal power plant is to be stored as latent heat in an insulated **solid** block of zinc which is maintained at its melting point of 420 °C. If the entire excess energy is used to melt zinc, calculate the minimum mass of solid zinc necessary for this purpose.

Specific latent heat of fusion of zinc is 1.15×10^5 J kg^{-1}.

(*b*) The temperature inside a certain outdoor closed storage room in a cold country is to be maintained at 0 °C when the outside temperature is at –30 °C. The room is thermally insulated with 20 cm thick concrete walls. The inner surfaces of the walls are in contact with a uniform water layer of sufficient thickness maintained at 0 °C as shown in the figure. Water is stirred internally to avoid formation of static frozen ice layers. (Assume that the stirring process does not add any heat to water.)



(i) Explain briefly how the temperature of the room can be maintained in 0 °C upto sometime using this method.

(ii) Calculate the minimum mass of the water layer which will ensure that the 0 °C temperature is maintained in the room upto 10 hours and only 25% of the mass of water is converted to ice during this time period.

Total mean surface area of all the walls is 120 m^2.
Thermal conductivity of concrete = 0.8 W m^{-1} °C^{-1}. Specific latent heat of fusion of ice = 3.35×10^5 J kg^{-1}.

(iii) Suppose the above mentioned entire water layer is frozen due to some unforseen reason and a uniform ice layer of thickness 5 cm is formed on the inner surface of concrete walls. Calculate the rate at which the heat from the 0 °C room begins to flow out as soon as the ice layer is formed. Thermal conductivity of ice = 2.2 W m^{-1} °C^{-1}. For calculations, assume that the total mean surface area of the ice layer through which the heat flows out is also 120 m^2.

(B) Radioisotope Thermoelectric Generators (RTGs) are used to generate electricity in space-crafts, satellites etc. An RTG consists of two subsystems.

(1) Thermal source:
It is a container of alpha particle emitting radioactive source. The kinetic energy produced by all the alpha particles is converted to thermal energy and absorbed by the container.

(2) Energy conversion system:
It is a thermoelectric generator which converts thermal energy absorbed by the container into electrical energy.

Consider an RTG of a certain space-craft which uses ^{238}Pu in the form of plutonium oxide (PuO_2) as the radioactive source. The radioactive source contains 2.38 kg of PuO_2 for which the fraction of ^{238}Pu in PuO_2 is 0.9 at the launch of the space-craft. The thermal energy absorbed per radioactive decay of ^{238}Pu by the container is 5.5 MeV. Half life of ^{238}Pu is 87.7 years and the corresponding decay constant is 0.0079 y^{-1} (= 2.5×10^{-10} s^{-1}). Avogadro number is 6.0×10^{23} atoms per mole.

(i) Find the initial activity in Bq of the radioisotope source at the launch of the space-craft.

(ii) If the efficiency of conversion of thermal power into electrical power is 7%, find the electrical power in the RTG at the launch of the space craft (1 MeV = 1.6×10^{-13} J).

(iii) Find the activity of the radioisotope source by the end of the 10 years mission of the space-craft. (Take $e^{-0.079} = 0.92$)

(iv) Find the electrical power produced by the RTG at the end of the mission.

(v) Find the percentage loss of the electrical power after the mission.

(vi) Give **one** advantage of using RTGs in space-crafts.

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